

Learning Objectives

- Explain the distinct roles of HTML, CSS, and JavaScript in building a web page.
- Describe how semantic HTML, responsive design, and frameworks improve a modern frontend.

Open any web page and you are seeing three technologies at work together. HTML builds the structure — the headings, paragraphs, and links, like the frame of a building. CSS adds the appearance — colors, layout, and fonts, like a building's interior. JavaScript adds behavior — search boxes that respond, buttons that count, screens that update. A modern frontend goes further: it makes pages meaningful, adaptable to every screen, and faster to build.

? Guiding question: How do HTML, CSS, and JavaScript work together to build a web page, and what makes a modern frontend meaningful, adaptable, and efficient to build?

Key Fact:
A web page is built from HTML (structure), CSS (appearance), and JavaScript (behavior); a modern frontend adds semantic HTML, responsive design, and frameworks.

Key Concepts:

- HTML / CSS / JavaScript
- semantic HTML
- accessibility / SEO
- responsive design
- framework

KEY TERMS
HTML — structure.
CSS — appearance.
JavaScript — behavior.



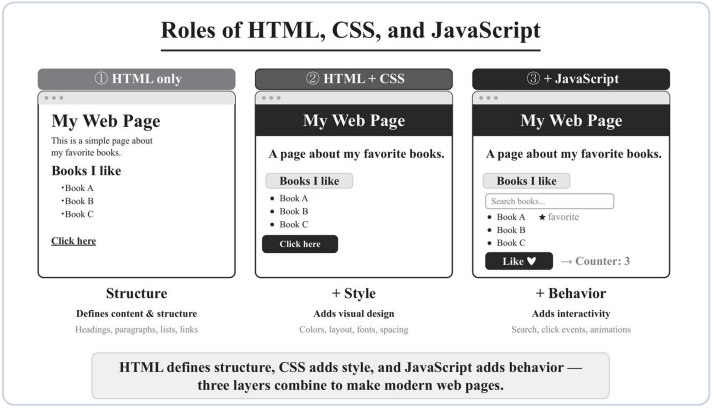
HTML structures, CSS styles, JS adds behavior.

Point!

1. The Roles of HTML, CSS, and JavaScript

(1) Web pages are mainly based on the following three technologies.

Technology	Role	Analogy
HTML	Defines the structure (skeleton) of a web page. Describes elements such as headings, paragraphs, images, and links.	The frame of a building
CSS	Specifies the appearance of a web page (colors, layout, size, fonts, etc.).	The interior and exterior of a building
JavaScript	Adds behavior and interactivity to a web page. Activates responses when buttons are pressed and updates only part of the screen.	The electrical and plumbing systems of a building



Roles of HTML, CSS, and JavaScript

(2) These three technologies each have independent roles, and a web page is created using all of them in combination.

2. Semantic HTML

(1) **(Semantic HTML)**…the concept of describing each part of a web page using element names (tags) that indicate their meaning.
e.g., Use <header> at the top of the page, <nav> for navigation, <main> for the body, and <footer> at the bottom of the page.

(2) Benefits of using semantic HTML

① Improved **(accessibility)**…screen reader software can correctly understand the structure of the page. Information becomes easier to convey to users with visual impairments.

② Improved **(SEO (Search Engine Optimization))**…search engines can more easily understand the content of a web page correctly, making it more likely to be displayed appropriately in search results.

3. Responsive Design

(1) **(Responsive design)**…a design concept in which the layout is automatically adjusted according to different screen sizes such as PCs, smartphones, and tablets.

(2) With a single web page, a display that is easy to view from any device can be achieved.

(3) As of 2026, websites are frequently accessed from smartphones, and support for responsive design is widely required. Against this backdrop, the **(mobile-first)** concept—prioritizing small screens such as smartphones from the design stage—has also become widespread.

4. Frameworks

(1) **(Framework)**…a structure that provides commonly used functions and components in advance, in order to make web application development more efficient.

(2) By using a framework, developers do not need to write all code from scratch and can develop more efficiently.

(3) Major frontend frameworks

Framework name	Characteristics
React	Developed by Meta (formerly Facebook). It can efficiently update only part of the screen and is suited to large-scale web applications.
Vue	Has a simple structure and allows web applications to be built incrementally from small features.
Next.js	In addition to React's features, includes mechanisms to speed up page display.

PAUSE & THINK

On a page you use often, which parts are structure (HTML), which are style (CSS), and which react to you (JavaScript)?



Responsive design re-flows to each screen size.

KEY TERMS

responsive design — layout adapts to screen size.
framework — ready-made functions for efficient development.



Frameworks assemble pages from ready-made parts.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A HTML is a technology that defines the structure (skeleton) of a web page.

B CSS is a technology that adds behavior to a web page.

C Responsive design is a concept in which the layout is automatically adjusted according to different screen sizes.

D Using a framework eliminates the need to write all code from scratch and allows efficient development.

(2) Match each role (a–c) with the most appropriate technology from the choices below (A–C).

a Defines the structure of a web page

b Specifies the appearance of a web page

c Adds behavior and interactivity to a web page

[Choices] **A** HTML **B** CSS **C** JavaScript

✓ Solution

(1)

A HTML defines the structure of a web page. Therefore, ○.

B CSS is a technology that specifies the appearance of a web page. Adding behavior is JavaScript. Therefore, ×.

C Correct definition of responsive design. Therefore, ○.

D Correct description of the benefit of frameworks. Therefore, ○.

(2)

a: Defining structure is HTML. **A**

b: Specifying appearance is CSS. **B**

c: Adding behavior is JavaScript. **C**

PAUSE & THINK

Why does a screen reader understand a page better when it uses <header>, <nav>, and <main> instead of plain boxes?

EXAM-STYLE QUESTION

Explain how HTML, CSS, and JavaScript each contribute to a library page that lists books, is styled, and filters the results as the user types. **[6] Refer to:** structure, appearance, behavior.

Try

1 Answer the following questions.

1. What is the four-letter abbreviation for the technology that defines the structure (skeleton) of a web page?
2. What is the three-letter abbreviation for the technology that specifies the appearance of a web page (colors, layout, size, etc.)?
3. What is the term for the technology that adds behavior and interactivity to a web page?
4. What is the term for the design concept in which the layout is automatically adjusted according to different screen sizes?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate benefit of using semantic HTML.

A Screen reader software can correctly understand the structure of the page.

B Search engines can more easily understand the content of a web page.

C The display speed of a web page necessarily becomes faster.

D Information becomes easier to convey to users with visual impairments.

(2) From the following options (A–D), choose the one that most appropriately describes a framework.

A A language for defining the structure of a web page

B A structure for making web application development more efficient

C A technology for encrypting data and sending it safely

D A design concept for supporting different screen sizes

PAUSE & THINK

Which of the benefits below is about how the page looks, and which is about how well others can find and read it?

Think as an Engineer — Investigate & Decide

A student club wants a simple web page that lists its events, is styled in the club's colors, and has a button showing how many people have signed up.

☐ **Write everything in one file**

☐ **Use HTML, CSS, and JavaScript for their proper roles**

1 · Observe first. Open one web page you use often and note two things on it that change when you click or type. Then, for three features of the club page — the list of events, the club's colors and layout, and the live sign-up counter — say which technology (HTML, CSS, or JavaScript) does each, and why.

2 · Make it reach everyone. Name one change that would make the page easier to read on a phone (responsive design), and one semantic-HTML choice that would help a screen reader.

3 · Decide. Recommend whether the club should separate HTML, CSS, and JavaScript, and give two reasons. **In pairs,** compare your role assignments and agree together on what the sign-up counter needs.

Give evidence for your answer.

Stuck? HTML holds the content; CSS styles it; JavaScript makes it react and update.

🌍 In a New Context

A news site must look good on phones and be found easily by search engines. Using what you learned, name one responsive-design choice and one semantic-HTML choice it should make, and say what each one improves.

❓ Lesson Question — Answered

The lesson asked how HTML, CSS, and JavaScript work together and what makes a modern frontend strong. A web page is built from three technologies with independent roles: HTML defines the structure, CSS specifies the appearance, and JavaScript adds behavior and interactivity — combined, they create the page. A modern frontend goes further: semantic HTML uses meaningful tags so screen readers and search engines understand the page (improving accessibility and SEO); responsive design adjusts the layout to any screen size, with a mobile-first approach; and frameworks such as React, Vue, and Next.js provide ready-made functions so developers build efficiently without writing everything from scratch. A modern web page is therefore not one technology but a combination — structured, styled, interactive, meaningful, adaptable, and efficient to build.

Exercise

KEY TERMS

accessibility — usable by people with impairments.
SEO — being found by search engines.

⭐ REFLECT & CHALLENGE

Reflect: Of structure, style, and behavior, which do you notice most when a web page is badly made, and why? **Challenge:** Name one website you use and describe one thing HTML, CSS, and JavaScript each do on it.

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Web pages are mainly composed of three technologies: (a), which defines the structure of the page; (b), which specifies the appearance; and (c), which adds behavior and interactivity. The concept of adjusting the layout according to different screen sizes such as PCs, smartphones, and tablets is called (d).

1. Fill in the blanks (a)–(d).

3 Answer the following questions.

(1) For each of tasks (a–f), indicate which technology is used to achieve it: **A** HTML, **B** CSS, **C** JavaScript.

a Describing the structure of headings and paragraphs

b Changing the color and size of text

c Updating part of the screen when a button is pressed

d Placing images and links

e Setting the background color of the entire page

f Validating the contents entered into a form

(2) From the following options (A–D), choose the one that most appropriately describes the reason responsive design is needed.

A To improve the display speed of websites

B Because websites are accessed from various devices such as PCs, smartphones, and tablets

C To raise rankings in search engines

D To add behavior to web pages

Key takeaway

Remember:
HTML structures, CSS styles, JavaScript behaves — and semantic HTML, responsive design, and frameworks make a modern frontend meaningful, adaptable, and efficient.